

Input

```
def gcd(a, b):  
    while b:  
        a, b = b, a % b  
    return a  
  
def mod_inverse(e, phi):  
    for d in range(1, phi):  
        if (e * d) % phi == 1:  
            return d  
  
p = int(input("Enter p: "))  
q = int(input("Enter q: "))  
e = int(input("Enter e: "))  
  
n = p * q  
phi = (p - 1) * (q - 1)  
  
d = mod_inverse(e, phi)  
  
print("n =", n)  
print("Private Key d =", d)
```

Output

```
Enter p: 61  
Enter q: 53  
Enter e: 17  
n = 3233  
Private Key d = 2753  
.
```